Reduced Optimal Control with PDE Constraint on an External Mesh: Model, Discretization, and First-Order Methods

Fachseminar

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Abstract

1 Problem Statement

We consider a linear-quadratic optimal control problem governed by the Poisson equation on a polygonal domain $\Omega \subset \mathbb{R}^2$ (in computations, typically $[0,1]^2$). A circular subregion $\omega \subset \Omega$ is tagged on the mesh and serves as the support of the control. The control acts as a volumetric source term restricted to ω:

$$-\Delta y = \chi_\omega \, u \quad \text{in } \Omega, \tag{1}$$

$$y = 0 \quad \text{on } \partial\Omega, \tag{2}$$

where χ_ω denotes the characteristic function of ω. We denote by $V = H_0^1(\Omega)$ the state space and by $U = L^2(\Omega)$ the control space. For a given target $y_d \in L^2(\Omega)$ and a regularization parameter $\beta > 0$, we minimize

$$J(y,u) = \tfrac{1}{2} \int_\Omega (y - y_d)^2 \, dx + \tfrac{\beta}{2} \int_\Omega u^2 \, dx. \tag{3}$$

The target is chosen as the pyramid-like profile

$$y_d(x) = 0.5 - \max\left(|x_1 - 0.5|, \, |x_2 - 0.5|\right), \tag{4}$$

which belongs to $L^2(\Omega)$. The control enters only on ω via χ_ω, so the effective source equals u on ω and vanishes on $\Omega \setminus \omega$. The PDE with homogeneous Dirichlet boundary conditions is well-posed for $u \in L^2(\Omega)$, yielding a unique weak solution $y \in V$ by Lax–Milgram.

2 Variational Formulation and Discretization

Weak formulation. Define the bilinear and linear forms

$$a(y,v) = \int_\Omega \nabla y \cdot \nabla v \, dx, \qquad b(u,v) = \int_\omega u \, v \, dx. \tag{5}$$

For any $u \in L^2(\Omega)$, the weak state equation reads: find $y \in V$ such that

$$a(y,v) = b(u,v) \quad \forall v \in V. \tag{6}$$

By coercivity of $a(\cdot,\cdot)$ on V and boundedness of $b(\cdot,\cdot)$, this admits a unique $y \in V$.

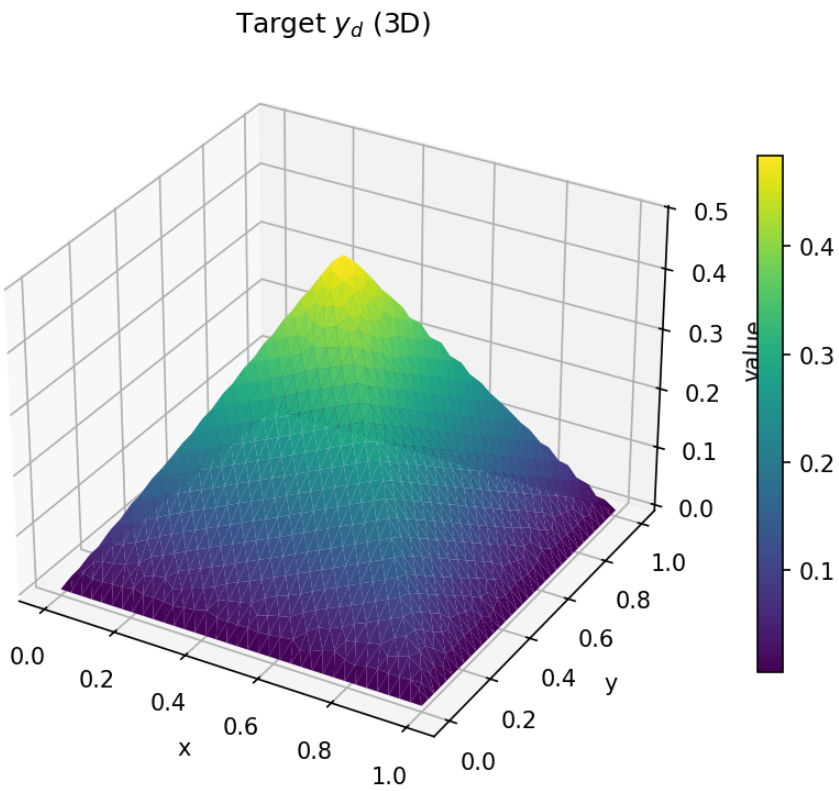

Figure 1: 3D surface of the target profile y_d on Ω.

Finite elements. Let $\mathcal{T}_h$ be a conforming triangulation of Ω. We use the conforming P1 space $V_h \subset V$ with nodal basis $\{\varphi_i\}_{i=1}^n$. The control is represented in the same P1 basis but measured with the L^2 inner product ("control metric"). We assemble the matrices

$$A_{ij} = \int_\Omega \nabla\varphi_j \cdot \nabla\varphi_i \, dx, \quad M_{ij} = \int_\Omega \varphi_j \, \varphi_i \, dx, \quad B_{ij} = \int_\omega \varphi_j \, \varphi_i \, dx, \quad (M_U)_{ij} = \int_\Omega \varphi_j \, \varphi_i \, dx. \tag{7}$$

Here, A is the stiffness matrix for state/adjoint solves, M is the state mass matrix used to weight the tracking term, B maps control coefficients to the state RHS via restriction to ω, and M_U is the control-space mass matrix defining the L^2 inner product in U_h.

Boundary conditions. Homogeneous Dirichlet boundary conditions are imposed on the state/adjoint by modifying A and zeroing the RHS at boundary degrees of freedom when solving. Importantly, no boundary conditions are applied to B or M_U; these operators remain the unmodified L^2-based forms on Ω.

Discrete state equation and objective. With coefficient vectors $\mathbf{y}, \mathbf{u} \in \mathbb{R}^n$ and target coefficients $\mathbf{y}_d \in \mathbb{R}^n$, the discrete state equation and cost read

$$A\,\mathbf{y} = B\,\mathbf{u}, \tag{8}$$

$$J(\mathbf{y}, \mathbf{u}) = \tfrac{1}{2}(\mathbf{y} - \mathbf{y}_d)^\top M(\mathbf{y} - \mathbf{y}_d) + \tfrac{\beta}{2}\mathbf{u}^\top MU\,\mathbf{u}. \tag{9}$$

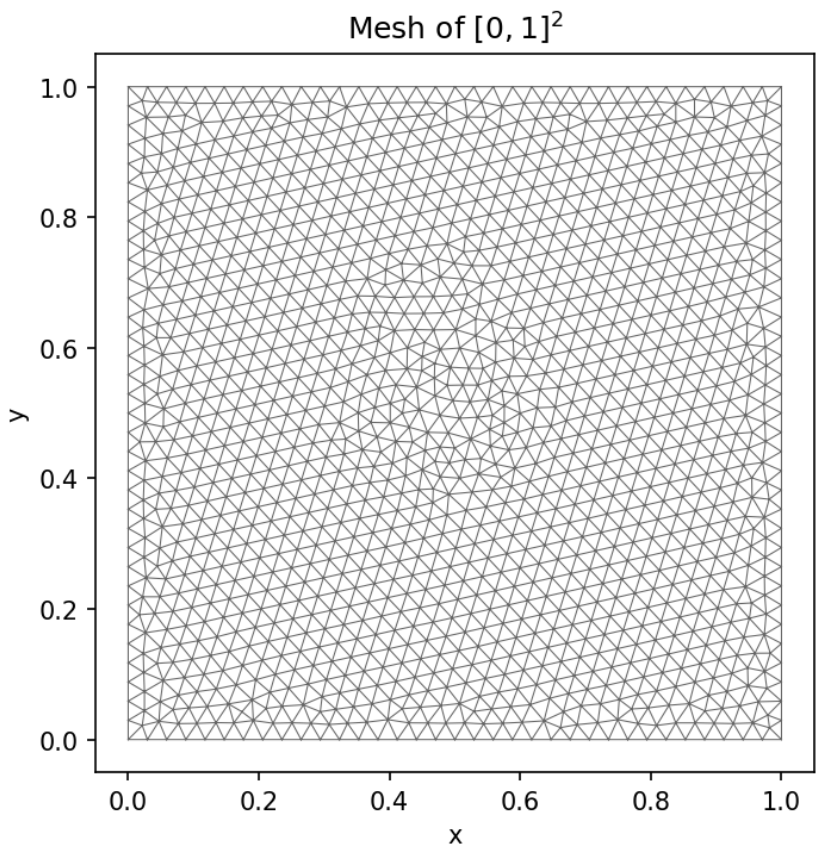

Figure 2: Triangular mesh of Ω used for the P1 finite element discretization.

3 Reduced Formulation

State elimination. Eliminating the state via the discrete PDE constraint gives $\mathbf{y}(\mathbf{u}) = A^{-1}B\,\mathbf{u}$ and the reduced functional $F\colon \mathbb{R}^n \to \mathbb{R}$:

$$F(\mathbf{u}) = \tfrac{1}{2}(A^{-1}B\,\mathbf{u} - \mathbf{y}_d)^\top M(A^{-1}B\,\mathbf{u} - \mathbf{y}_d) + \tfrac{\beta}{2}\,\mathbf{u}^\top MU\,\mathbf{u}. \tag{10}$$

Adjoint and gradient (continuous view). Introducing an adjoint variable $p \in V$ and the Lagrangian

$$\mathcal{L}(y, u, p) = \tfrac{1}{2}\|y - y_d\|^2_{L^2(\Omega)} + \tfrac{\beta}{2}\|u\|^2_{L^2(\Omega)} + a(y, p) - b(u, p), \tag{11}$$

the optimality system reads: state $a(y, v) = b(u, v)$, adjoint $a(v, p) = (y - y_d, v)$ for all $v \in V$, and variational gradient condition $(\beta u + \chi_\omega p, w)_{L^2(\Omega)} = 0$ for all $w \in U$. Hence, formally, $\beta u + p\,\chi_\omega = 0$ in Ω.

Adjoint and gradient (discrete view). In coefficients, the adjoint solves

$$A\,\mathbf{p} = M\,(\mathbf{y}(\mathbf{u}) - \mathbf{y}_d), \tag{12}$$

and the gradient with respect to the control space inner product induced by MU is

$$\nabla F(\mathbf{u}) = MU^{-1}\,B^\top\mathbf{p} + \beta\,\mathbf{u}. \tag{13}$$

The first-order optimality condition $\nabla F(\mathbf{u}_*) = 0$ is equivalent to

$$B^\top\mathbf{p}_* + \beta\,MU\,\mathbf{u}_* = 0, \qquad A\,\mathbf{p}_* = M\,(\mathbf{y}(\mathbf{u}_*) - \mathbf{y}_d). \tag{14}$$

Hessian, convexity, and Lipschitz constant. The reduced Hessian applied to $\mathbf{v}$ is

$$H\,\mathbf{v} = MU^{-1}\,B^\top A^{-1} M A^{-1} B\,\mathbf{v} + \beta\,\mathbf{v}. \tag{15}$$

Since A, M, and MU are symmetric positive definite on their natural subspaces and B is linear, the quadratic form associated with H is strictly positive for $\beta > 0$; thus, F is strictly convex with a unique minimizer. In the U-metric, the gradient is Lipschitz with constant

$$L = \left\| MU^{-1/2} B^\top A^{-1} M A^{-1} B\, MU^{-1/2} \right\|_2 + \beta, \tag{16}$$

namely the spectral radius of the SPD term plus β.